Uniform Inference with Jointly Weak Variance and Fourth-Moment Features

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10 October 2026

C32-80 JEL C32 Source OP-0603

Abstract

We study a declared nonparametric experiment with independent observations from known volatility regimes, a common nonsingular impact matrix, conditionally independent shocks, and bounded eighth moments. Variance contrasts and baseline fourth cumulants may weaken jointly. A global inverse inequality yields uniformly honest confidence sets for the full impact matrix modulo signed permutations. We prove matching worst-case expected-diameter bounds at every confidence level, including complete feature failure and arbitrary weakening sequences. A single procedure also attains the rate without knowing the strength lower bound. We give matching squared-error bounds and state separately the conditions for labelled matrices, other responses, and dimension one.

1 Declared experiment, target, and loss

Fix integers $d \geq 2$ and $S \geq 0$, and constants

$$0 < b_- < b_+ < \infty, \qquad 0 < v_- < 1 \leq v_+ < \infty, \qquad M_8 \geq 105, \qquad 0 < \nu \leq \frac{1}{S+1}.$$

All constants, including d and S, remain fixed as the sample size increases. There are $S+1$ known regimes, numbered $0, \ldots, S$. For each admissible sample size T, their known deterministic counts satisfy

$$\sum_{s=0}^{S} T_s = T, \qquad T_s \geq \nu T.$$

Observations are independent across all indices. Within regime s, they have the common distribution of $U = BE_s$. The coordinates of E_s are mutually independent, have mean zero, have variances $d_{si} \in [v_-, v_+]$, and satisfy

$$\mathbb{E}|E_{si}|^8 \leq M_8.$$

In regime 0, $d_{0i} = 1$ for every i. The matrix B is common across regimes and every singular value lies in $[b_-, b_+]$. Apart from these conditions, the coordinate shock laws are arbitrary and unknown; the experiment imposes no density or parametric shape assumption. The laws and B may depend on T. The observed data O_T are the known-regime innovation vectors U, not estimated residuals from an additional, unspecified time-series model.

Write $D_s = \operatorname{diag}(d_{s1}, \ldots, d_{sd})$, $c_i = \mathbb{E}E_{0i}^4 - 3$, and set

$$\delta(D, c)^2 = \min_{i<j} \left\{ \sum_{s=1}^{S} (d_{si} - d_{sj})^2 + c_i^2 + c_j^2 \right\}. \tag{1.1}$$

The sum is zero when $S = 0$. This index describes the specified moment features, not every possible form of non-Gaussian identification. Let $\mathcal{M}_T(k)$ be the structural models just declared with $\delta(D, c) \geq k$. The nonnegative k indexes the benchmark class. The first confidence construction uses k; Section 6A gives a single procedure without it. Below, $k \in [0, \bar{k}]$, for a sufficiently small fixed $\bar{k} > 0$ specified in Section 7. This interval includes arbitrary rates of convergence to zero and the sequence identically zero.

Let $\mathcal{G}$ be the finite group of signed permutation matrices of order d. The primary response is the entire impact matrix modulo its column signs and labels:

$$\tau(m) = [B], \qquad d_{\mathcal{G}}([B], [B']) = \min_{P \in \mathcal{G}} \left\| B - B'P \right\|_{\mathrm{F}}.$$

This is a quotient metric: the action is isometric, the minimum is attained because $\mathcal{G}$ is finite, and composition of minimizing group elements proves the triangle inequality. Denote the compact quotient target space by $\mathcal{X}$. Its diameter is at most

$$D_{\max} = 2\sqrt{d}\, b_+.$$

A confidence procedure is a measurable random closed subset C_T of $\mathcal{X}$; independent auxiliary randomization is allowed. Set $\operatorname{diam}(\varnothing) = 0$. For $0 < \alpha < 1$, define

$$\mathcal{R}_T(k, \alpha) = \inf_{C} \sup_{m \in \mathcal{M}_T(k)} \mathbb{E}_m \operatorname{diam}(C),$$

where the infimum ranges over procedures satisfying

$$\inf_{m \in \mathcal{M}_T(k)} \mathbb{P}_m\{[B_m] \in C\} \geq 1 - \alpha.$$

The full space $\mathcal{X}$ is one such procedure. Define the rate

$$r(T, k) = \begin{cases} \min\{1, (\sqrt{T}\, k)^{-1}\}, & k > 0, \\ 1, & k = 0. \end{cases} \tag{1.2}$$

Theorem 1 (Full quotient-matrix response)**.** *There is $\bar{k} > 0$, depending only on the displayed class constants, such that for each $0 < \alpha < 1$ there are constants $0 < c_\alpha \leq C_\alpha < \infty$ with*

$$c_\alpha r(T, k) \leq \mathcal{R}_T(k, \alpha) \leq C_\alpha r(T, k) \tag{1.3}$$

for every admissible T and every $k \in [0, \bar{k}]$. The upper procedure has finite-sample coverage at least $1 - \alpha$; its diameter bound holds deterministically for every data set. It therefore holds in worst-case expectation without an α-sized remainder. At $k = 0$, the worst-case expected diameter is bounded away from zero.

For any sequence $k_T \in [0, \bar{k}]$, every procedure sequence satisfying

$$\liminf_{T \to \infty} \inf_{m \in \mathcal{M}_T(k_T)} \mathbb{P}_m\{[B_m] \in C_T\} \geq 1 - \alpha$$

obeys

$$\liminf_{T \to \infty} \frac{\sup_{m \in \mathcal{M}_T(k_T)} \mathbb{E}_m \operatorname{diam}(C_T)}{r(T, k_T)} \geq c'_\alpha > 0. \tag{1.4}$$

The finite-sample upper procedure also attains this asymptotic rate.

The finite-T infimum is deliberately over finite-T honest procedures. Taking a finite-T infimum over merely asymptotically honest sequences would be invalid. Labelled matrices, response maps, covariance responses, and $d = 1$ are treated in Sections 11–13. No universal rate is asserted for arbitrary response functionals without stated regularity and nonflatness.

2 Raw features and a uniform concentration bound

Use Frobenius norms on matrices and fourth-order tensors, including all ordered tensor entries. Give a tuple the Euclidean sum-of-squares norm. For a symmetric matrix Σ, define its Wick tensor by

$$W(\Sigma)_{abcd} = \Sigma_{ab}\Sigma_{cd} + \Sigma_{ac}\Sigma_{bd} + \Sigma_{ad}\Sigma_{bc}.$$

For $\vartheta = (B, D_1, \dots, D_S, c)$, let

$$\begin{aligned} \Sigma_0 &= BB^{\mathsf{T}}, & \Sigma_s &= BD_sB^{\mathsf{T}} \quad (1 \le s \le S), \\ K_{\mathrm{raw}} &= \sum_i c_i b_i^{\otimes 4}, & M_4 &= K_{\mathrm{raw}} + W(\Sigma_0), \\ F(\vartheta) &= (\Sigma_0, \dots, \Sigma_S, M_4), \end{aligned}$$

where b_i is column i of B. Zero means and coordinate independence give the second-moment formulas. For the fourth-moment formula, expand a product of four coordinates of BE_0. Independence and zero means annihilate every term with a coordinate appearing once. Pair-pair terms give $W(\Sigma_0)$; four-equal-coordinate terms leave coefficient $\mathbb{E}E_{0i}^4 - 3$. Thus the formula holds for nonsymmetric shock laws and in the presence of nonzero third moments.

Let $\widehat{F}$ consist of the empirical raw second moment in each regime and the empirical raw fourth moment in regime 0. These moments are not centered by a sample mean: the experiment imposes zero population means, so the raw statistics are unbiased.

Put $H_{\mathrm{obs},8} = b_+^8 d^4 M_8$. For every regime,

$$\begin{aligned} \mathbb{E}\,\|U\|^8 &\le b_+^8\,\mathbb{E}\left(\sum_i E_{si}^2\right)^4 \\ &\le b_+^8 d^3 \sum_i \mathbb{E}|E_{si}|^8 \le H_{\mathrm{obs},8}. \end{aligned}$$

Here $(\sum_i a_i)^4 \le d^3 \sum_i a_i^4$ for $a_i \ge 0$ follows from Hölder's inequality. Consequently $\mathbb{E}\,\|U\|^4 \le \sqrt{H_{\mathrm{obs},8}}$. Each entry U_aU_b has variance at most $\sqrt{H_{\mathrm{obs},8}}$; each entry $U_aU_bU_cU_e$ has variance at most $H_{\mathrm{obs},8}$. Independence across observations and $T_s \ge \nu T$ therefore imply

$$\mathbb{E}_m \left\| \widehat{F} - F(\vartheta_m) \right\|^2 \le \frac{V}{T}, \qquad V = \nu^{-1}\left[(S+1)d^2\sqrt{H_{\mathrm{obs},8}} + d^4 H_{\mathrm{obs},8}\right]. \tag{2.1}$$

Correlations between feature entries are harmless: the expected squared Euclidean norm is the sum of the individual second moments. For the confidence radius

$$R_T = \frac{\sqrt{V/\alpha}}{\sqrt{T}}, \tag{2.2}$$

Markov's inequality gives

$$\sup_m \mathbb{P}_m\{\left\| \widehat{F} - F(\vartheta_m) \right\| > R_T\} \le \alpha. \tag{2.3}$$

3 A compact outer parameter set

The true cumulants satisfy

$$-2 \le c_i \le \sqrt{M_8} - 3,$$

because $\mathbb{E}E_{0i}^4 \ge 1$ and $\mathbb{E}E_{0i}^4 \le \sqrt{\mathbb{E}E_{0i}^8}$. Use the larger compact box $|c_i| \le K$, where $K = 3 + \sqrt{M_8}$. Let Θ_k contain the tuples $\vartheta = (B, D, c)$ satisfying the singular-value and diagonal-variance

bounds of Section 1, this cumulant box, and $\delta(D, c) \geq k$. Every true parameter in $\mathcal{M}_T(k)$ belongs to Θ_k. The set is compact and is specified by finitely many polynomial inequalities: positive semidefiniteness of $B^\mathsf{T}B - b_-^2 I$ and $b_+^2 I - B^\mathsf{T}B$, the diagonal boxes, and

$$\sum_s (d_{si} - d_{sj})^2 + c_i^2 + c_j^2 \geq k^2 \quad \text{for every } i < j.$$

Define

$$\begin{aligned} A_T(\widehat{F}) &= \{\vartheta \in \Theta_k : \left\| F(\vartheta) - \widehat{F} \right\| \leq R_T\}, \\ C_T &= \{[B] : (B, D, c) \in A_T(\widehat{F})\}. \end{aligned} \tag{3.1}$$

Candidates in Θ_k need not be realizable by actual independent shock laws. This outer enlargement is intentional: the confidence set need only contain the true structural parameter, and all deterministic inverse bounds below hold for every candidate.

By (2.3), the true tuple belongs to A_T with probability at least $1 - \alpha$. Thus C_T is uniformly honest, including at $k = 0$. It may be empty on some data sets; the specified zero-diameter convention applies. The construction never whitens the empirical covariance and remains defined when that covariance is singular or ill-conditioned.

For completeness, these sets are measurable. Their graph in $(\widehat{F}, \vartheta)$ is closed and the parameter coordinate lies in a fixed compact set. Projection after additionally restricting $[B]$ to any closed target set remains closed. Every open set in a compact metric space is a countable union of closed balls lying inside it; consequently the event that C_T hits an open target set is Borel. For $a > 0$, the set of feature vectors with $\operatorname{diam}(C_T) \geq a$ is also closed: introduce two candidate tuples, impose $\min_{P \in \mathcal{G}} \|B - B'P\|_{\mathrm{F}} \geq a$, and use compact projection. Strict events follow by countable unions. Composing with the polynomial empirical moment map proves measurability. The constraints are explicit finite algebraic constraints; this statistical theorem asserts neither efficient numerical optimization nor exact computation for arbitrary unencoded real inputs.

4 Uniformly Lipschitz whitening

For two candidate tuples, put $e = \|F(\vartheta) - F(\vartheta')\|$, and let

$$H = \Sigma_0^{1/2}, \quad H' = (\Sigma_0')^{1/2}, \qquad L = H^{-1}, \quad L' = (H')^{-1},$$

using principal positive definite square roots. The eigenvalues of H, H' lie in $[b_-, b_+]$. For $X = H - H'$,

$$HX + XH' = \Sigma_0 - \Sigma_0'.$$

Taking Frobenius inner products with X gives

$$2b_- \|X\|_{\mathrm{F}}^2 \leq \langle X, HX + XH' \rangle \leq \|X\|_{\mathrm{F}} \left\| \Sigma_0 - \Sigma_0' \right\|_{\mathrm{F}}.$$

Indeed, the two inner-product terms are $\operatorname{tr}(X^\mathsf{T}HX)$ and $\operatorname{tr}(X^\mathsf{T}XH')$, each at least $b_- \|X\|_{\mathrm{F}}^2$. Hence, also when $X = 0$,

$$\left\| H - H' \right\|_{\mathrm{F}} \leq \frac{\|\Sigma_0 - \Sigma_0'\|_{\mathrm{F}}}{2b_-}, \qquad \left\| L - L' \right\|_{\mathrm{F}} \leq \frac{\|\Sigma_0 - \Sigma_0'\|_{\mathrm{F}}}{2b_-^3}. \tag{4.1}$$

The second bound uses $L - L' = H^{-1}(H' - H)(H')^{-1}$.

The whitened features are

$$A_s = L\Sigma_s L, \quad K_{\mathrm{w}} = L^{\otimes 4}(M_4 - W(\Sigma_0)), \qquad F_{\mathrm{w}}(\vartheta) = (A_1, \ldots, A_S, K_{\mathrm{w}}).$$

Writing $Q = LB$, we have $QQ^{\mathsf{T}} = I$, so Q is orthogonal and

$$A_s = QD_sQ^{\mathsf{T}}, \qquad K_{\mathrm{w}} = \sum_i c_i q_i^{\otimes 4}. \tag{4.2}$$

Candidate covariances obey $\|\Sigma_0\|_{\mathrm{F}} \le \sqrt{d}\, b_+^2$ and $\|\Sigma_s\|_{\mathrm{op}} \le v_+ b_+^2$. Telescoping $L\Sigma_s L$ and applying (4.1) yield

$$\left\|A_s - A_s'\right\|_{\mathrm{F}} \le C_A e, \qquad C_A = b_-^{-2} + v_+ b_+^2 b_-^{-4}.$$

Each of the three summands of $W(\Sigma)$ rearranges $\Sigma \otimes \Sigma$, so

$$\begin{aligned}\left\|W(\Sigma_0) - W(\Sigma_0')\right\|_{\mathrm{F}} &\le 3(\|\Sigma_0\|_{\mathrm{F}} + \left\|\Sigma_0'\right\|_{\mathrm{F}}) \left\|\Sigma_0 - \Sigma_0'\right\|_{\mathrm{F}} \\ &\le 6\sqrt{d}\, b_+^2 e.\end{aligned}$$

Also $\|K_{\mathrm{raw}}'\|_{\mathrm{F}} \le \sum_i |c_i'|\, \|b_i'\|^4 \le dK b_+^4$. The four-factor telescoping identity gives

$$\left\|L^{\otimes 4} - (L')^{\otimes 4}\right\|_{\mathrm{op}} \le 4 b_-^{-3} \left\|L - L'\right\|_{\mathrm{op}}.$$

Consequently

$$\left\|K_{\mathrm{w}} - K_{\mathrm{w}}'\right\|_{\mathrm{F}} \le C_K e, \qquad C_K = b_-^{-4}(1 + 6\sqrt{d}\, b_+^2) + 2dK b_+^4 b_-^{-6}.$$

We have proved

$$\left\|F_{\mathrm{w}}(\vartheta) - F_{\mathrm{w}}(\vartheta')\right\| \le L_{\mathrm{w}} \left\|F(\vartheta) - F(\vartheta')\right\|, \qquad L_{\mathrm{w}} = \sqrt{SC_A^2 + C_K^2}. \tag{4.3}$$

The constants depend only on the declared compact bounds. No local approximation, eigenvector selection, or eigenvalue-gap assumption has been used.

5 The global joint inverse inequality

For diagonal matrices $D_s = \mathrm{diag}(d_{s1}, \dots, d_{sd})$, tensor $K_0 = \sum_i c_i e_i^{\otimes 4}$, and $R \in O(d)$, put $P_{ij} = R_{ij}^2$. The matrix P is doubly stochastic. Let $\mathcal{E}(R)^2$ be the sum of the squared distances of $R^{\mathsf{T}} D_s R$ from the diagonal matrix subspace and the squared distance of K_0, expressed in the R basis, from $\mathrm{span}\{e_j^{\otimes 4}\}$.

For each covariance feature, its off-diagonal squared norm is

$$\sum_i d_{si}^2 - \sum_j \left(\sum_i d_{si} P_{ij}\right)^2 = \sum_{i<\ell} (d_{si} - d_{s\ell})^2 \sum_j P_{ij} P_{\ell j}. \tag{5.1}$$

To verify this, use $\sum_j P_{ij} = 1$ and apply in each column the weighted-variance identity

$$\sum_i w_i x_i^2 - \left(\sum_i w_i x_i\right)^2 = \sum_{i<\ell} w_i w_\ell (x_i - x_\ell)^2, \qquad \sum_i w_i = 1.$$

The diagonal fourth-tensor coefficients in the R basis are

$$\beta_j = \sum_i c_i R_{ij}^4 = \sum_i c_i P_{ij}^2.$$

Orthogonal changes of basis preserve the tensor Frobenius norm, whose square is $\sum_i c_i^2$. Cauchy–Schwarz and $\sum_i P_{ij}^2 \le \sum_i P_{ij} = 1$ imply

$$\beta_j^2 \le \left(\sum_i c_i^2 P_{ij}^2\right)\left(\sum_i P_{ij}^2\right) \le \sum_i c_i^2 P_{ij}^2.$$

Its squared residual is therefore at least

$$\sum_i c_i^2 \left(1 - \sum_j P_{ij}^2\right) = \sum_{i<\ell} (c_i^2 + c_\ell^2) \sum_j P_{ij} P_{\ell j}. \tag{5.2}$$

No sign restriction on c_i is used. Combining (5.1)–(5.2) gives

$$\mathcal{E}(R)^2 \geq \delta(D,c)^2 \sum_{i<\ell} \sum_j P_{ij} P_{\ell j} = \frac{\delta(D,c)^2}{2} \left(d - \sum_{ij} P_{ij}^2\right). \tag{5.3}$$

Every doubly stochastic matrix is a convex combination of permutation matrices, as the following elementary proof shows. In the positive-entry bipartite graph, the neighbor set $N(I)$ of any set I of rows satisfies

$$|I| = \sum_{i \in I, j} P_{ij} \leq \sum_{j \in N(I)} \sum_i P_{ij} = |N(I)|.$$

This condition gives a perfect matching. Indeed, take a maximum matching. An unmatched row generates an alternating search tree. Without an augmenting path, every reached column is matched and its matched row is reached. The reached rows then outnumber the reached columns while all their neighbors are reached, contradicting the displayed inequality. Thus there is a permutation π whose selected entries are positive. Subtract λ times its permutation matrix, where λ is the smallest selected entry. If $\lambda < 1$, divide the residual by $1 - \lambda$; it is doubly stochastic and has fewer positive entries. If $\lambda = 1$, the original matrix is that permutation matrix. Induction on the number of positive entries proves the decomposition.

Apply the decomposition to P and take its inner product with P. Some permutation π satisfies

$$\sum_i P_{i,\pi(i)} \geq \sum_{ij} P_{ij}^2.$$

Choose signs matching the entries $R_{i,\pi(i)}$. Since $|r| \geq r^2$ when $|r| \leq 1$,

$$\begin{aligned} \min_{P_0 \in \mathcal{G}} \|R - P_0\|_{\mathrm{F}}^2 &= 2d - 2 \max_\pi \sum_i |R_{i,\pi(i)}| \\ &\leq 2 \left(d - \sum_{ij} P_{ij}^2\right). \end{aligned}$$

Together with (5.3), this proves, for $\delta(D,c) > 0$,

$$\min_{P_0 \in \mathcal{G}} \|R - P_0\|_{\mathrm{F}} \leq \frac{2\mathcal{E}(R)}{\delta(D,c)}. \tag{5.4}$$

For two tuples of the form (4.2), put $R = Q^{\mathsf{T}} Q'$. Express both in the Q' basis and project onto the off-diagonal matrix and fourth-tensor subspaces. The competing tuple, with its own arbitrary diagonal D'_s and c'_i, is annihilated. Orthogonal projection is a contraction, so

$$\mathcal{E}(Q^{\mathsf{T}} Q') \leq \left\| F_{\mathrm{w}}(\vartheta) - F_{\mathrm{w}}(\vartheta') \right\|.$$

Equation (5.4) therefore yields

$$\min_{P_0 \in \mathcal{G}} \left\| Q - Q' P_0 \right\|_{\mathrm{F}} \leq \frac{2 \left\| F_{\mathrm{w}}(\vartheta) - F_{\mathrm{w}}(\vartheta') \right\|}{\delta(D,c)}. \tag{5.5}$$

The two signed-permutation minimizations agree by orthogonal multiplication and replacing a group element by its inverse. This is a global inequality allowing different nuisance variances and cumulants.

6 Deterministic confidence diameter and the upper theorem

Reconstruct $B = HQ$ and $B' = H'Q'$. Equations (4.1), (4.3), and (5.5) imply

$$\begin{aligned} d_{\mathcal{G}}([B],[B']) &\le \left\|H - H'\right\|_{\mathrm{F}} + b_+ \min_{P_0 \in \mathcal{G}} \left\|Q - Q'P_0\right\|_{\mathrm{F}} \\ &\le \left[\frac{1}{2b_-} + \frac{2b_+ L_{\mathrm{w}}}{\delta(D,c)}\right] \left\|F(\vartheta) - F(\vartheta')\right\|. \end{aligned} \tag{6.1}$$

Restrict $\bar{k} \le 1$. For any $\vartheta, \vartheta' \in \Theta_k$ with $k > 0$, this gives

$$d_{\mathcal{G}}([B],[B']) \le \frac{L_I}{k} \left\|F(\vartheta) - F(\vartheta')\right\|, \qquad L_I = \frac{1}{2b_-} + 2b_+ L_{\mathrm{w}}. \tag{6.2}$$

Any two accepted candidates in A_T have raw-feature distance at most $2R_T$. Hence, on every data set, not merely on a coverage event,

$$\operatorname{diam}(C_T) \le \min\left\{D_{\max}, \frac{2L_I\sqrt{V/\alpha}}{\sqrt{T}\,k}\right\} \quad (k > 0). \tag{6.3}$$

At $k = 0$, $\operatorname{diam}(C_T) \le D_{\max}$. Empty sets satisfy the same bounds. For $x \ge 0$, $\min\{D_{\max}, Ax\} \le \max\{D_{\max}, A\}\min\{1, x\}$. Together with finite-sample coverage from Section 3, (6.3) proves the upper bound in Theorem 1, with

$$C_\alpha = \max\{D_{\max},\, 2L_I\sqrt{V/\alpha}\}.$$

Identification from these raw moments is an immediate corollary: if two candidates have identical raw moments and the first has $\delta > 0$, then (6.1) forces identical quotient impact matrices. This does not assert that all distributional identification fails whenever $\delta = 0$.

6A. One confidence procedure without the strength bound

Define (3.1) using Θ_0 for every sample, without inputting k, and call its quotient image C_T^{adapt}. The same concentration event gives coverage at least $1 - \alpha$ uniformly over the entire declared structural class, including $\delta = 0$.

Fix any true tuple ϑ_* with $\delta_* > 0$, and put

$$e_* = \left\|\widehat{F} - F(\vartheta_*)\right\|.$$

Every accepted candidate ϑ satisfies

$$\|F(\vartheta) - F(\vartheta_*)\| \le R_T + e_*.$$

The bound (6.1) uses only the separation of its first tuple. Apply it with the true tuple first, even if ϑ_* is not accepted by the confidence set. Comparing each of two accepted candidates to the true target and using the triangle inequality gives, for every data set, with $L_0 = (2b_-)^{-1}$ and $L_1 = 2b_+L_{\mathrm{w}}$,

$$\operatorname{diam}(C_T^{\mathrm{adapt}}) \le \min\left\{D_{\max}, 2\left[L_0 + \frac{L_1}{\delta_*}\right](R_T + e_*)\right\}. \tag{6.4}$$

The candidates themselves need not have positive δ; empty sets again satisfy the bound. By (2.1), $\mathbb{E}e_* \le \sqrt{V/T}$. Taking expectations of each of the two bounds in (6.4) yields

$$\mathbb{E}\operatorname{diam}(C_T^{\mathrm{adapt}}) \le \min\left\{D_{\max}, \frac{2[L_0 + L_1/\delta_*]\,[\sqrt{V/\alpha} + \sqrt{V}]}{\sqrt{T}}\right\}. \tag{6.5}$$

For $\delta_* \geq k > 0$ and $k \leq \bar{k} \leq 1$, this is at most a fixed multiple of $r(T,k)$; for $k = 0$, use $D_{\max}$. Thus a single procedure, without knowledge of k or δ_*, attains the upper minimax expected-diameter rate on every $\mathcal{M}_T(k)$. Its every-data-set bound is (6.4), which contains the realized moment error. The fixed numerical bound (6.3) belongs to the k-restricted procedure.

7 A non-Gaussian hard family inside the nonparametric class

Let φ be the $N(0,1)$ density and define

$$\begin{aligned} H_4(z) &= z^4 - 6z^2 + 3, \\ H_8(z) &= z^8 - 28z^6 + 210z^4 - 420z^2 + 105, \\ h(x) &= e^{-x^2}\left[H_4(\sqrt{3}\,x) - \frac{1}{4}H_8(\sqrt{3}\,x)\right]. \end{aligned}$$

The functions h, h' are bounded because they are polynomials times e^{-x^2}. Put $M_h = \|h\|_\infty > 0$. For $r = 0, 2, 4, 8$, the change of variable $y = \sqrt{3}\,x$ gives

$$\mathbb{E}_\varphi[X^r h(X)] = 3^{-(r+1)/2}\mathbb{E}\left[Y^r\left(H_4(Y) - \frac{1}{4}H_8(Y)\right)\right], \qquad Y \sim N(0,1).$$

For $j = 4, 8$, direct differentiation gives $H_j(y)\varphi(y) = (-1)^j\varphi^{(j)}(y)$. Repeated integration by parts, with vanishing boundary terms because a Gaussian dominates each polynomial, therefore yields

$$\mathbb{E}[Y^r H_j(Y)] = \mathbb{E}\left[\frac{d^j}{dy^j}y^r\Big|_{y=Y}\right].$$

Consequently

$$\begin{aligned} \mathbb{E}_\varphi h = 0, \qquad & \qquad \mathbb{E}_\varphi X^2 h = 0, \\ A := \mathbb{E}_\varphi X^4 h = \frac{24}{9\sqrt{3}} = \frac{8}{3\sqrt{3}} > 0, & \\ \mathbb{E}_\varphi X^8 h = \frac{5040 - 40320/4}{81\sqrt{3}} = -\frac{5040}{81\sqrt{3}}. & \end{aligned} \tag{7.1}$$

For example, $\mathbb{E}[Y^8 H_4(Y)] = (8!/4!)\mathbb{E}Y^4 = 5040$ and $\mathbb{E}[Y^8 H_8(Y)] = 8! = 40320$, verifying the numerical constants.

For $0 \leq \varepsilon \leq (2M_h)^{-1}$, the function

$$f_\varepsilon(x) = \varphi(x)(1 + \varepsilon h(x))$$

is a strictly positive symmetric density between $\varphi/2$ and $3\varphi/2$. Equation (7.1) shows that it has mean zero, variance one, fourth cumulant $A\varepsilon$, and eighth moment $105 - 5040\varepsilon/(81\sqrt{3})$. In particular,

$$\mathrm{cum}_4(f_{k/A}) = k, \qquad \mathbb{E}_{f_{k/A}}X^8 = 105 - \frac{70}{3}k \leq 105. \tag{7.2}$$

For $k > 0$, the nonzero fourth cumulant proves non-Gaussianity.

Choose henceforth

$$\bar{k} = \min\left\{1, \frac{1}{10d}, \frac{1 - v_-}{2d}, \frac{A}{2M_h}\right\} > 0. \tag{7.3}$$

For $k \in [0, \bar{k}]$, let $Z_1, \ldots, Z_d$ be independent with density $f_{k/A}$. In the baseline regime set $E_0 = Z$. If $S \geq 1$, use

$$D_1 = \mathrm{diag}(1 - k, 1 - 2k, \ldots, 1 - dk), \qquad E_1 = D_1^{1/2}Z,$$

and $D_s = I, E_s = Z$ for $s \geq 2$, with independent copies for all observations. When $S = 0$, there are no other regimes. Every variance lies both in $[0.9, 1]$ and in $[v_-, v_+]$. Every coordinate eighth moment is at most 105, because the scale factors are at most one. This family therefore meets the same moment cap $M_8 \geq 105$ as the upper theorem. Its baseline cumulants are all k, and

$$\delta(D, c) = \begin{cases} \sqrt{2}\, k, & S = 0, \\ \sqrt{3}\, k, & S \geq 1. \end{cases}$$

For the second case, adjacent coordinates minimize $(i - j)^2 k^2 + 2k^2$. Thus the family lies in $\mathcal{M}_T(k)$, including at $k = 0$. When $S \geq 1$, both nontrivial variance contrasts and baseline fourth cumulants vanish with k.

8 Quantitative closeness of the rotation experiments

Put $b_0 = (b_- + b_+)/2$. Let R_θ rotate the first two coordinates through angle θ and act as the identity on the others. Set $B_\theta = b_0 R_\theta$, initially for $|\theta| \leq \theta_0 = 1/8$. Every singular value equals b_0, so the general spectral bounds are respected; they need not contain 1. Let $P_{\theta,s}$ be the one-observation law in regime s with the shocks of Section 7. A common rescaling $x = b_0 y$ leaves chi-square divergence unchanged. In the y coordinates, the density is

$$\begin{aligned} p_{\theta,s}(y) &= g_{\theta,s}(y) H_{\theta,s}(y), \\ g_{\theta,s} &= \text{density of } N(0, R_\theta D_s R_\theta^\mathsf{T}), \\ H_{\theta,s}(y) &= \prod_i \left[1 + \frac{k}{A} h(z_i)\right], \qquad z = D_s^{-1/2} R_\theta^\mathsf{T} y. \end{aligned} \tag{8.1}$$

Each factor of $H_{\theta,s}$ lies in $[1/2, 3/2]$. Write $a = 0.9$, and let φ_a and φ_1 be the $N(0, aI_d)$ and $N(0, I_d)$ densities. Since every eigenvalue of D_s lies in $[a, 1]$, direct comparison of Gaussian exponents and normalizing constants gives

$$\begin{aligned} &g_{\theta,s}(y) \geq a^{d/2} \varphi_a(y), \quad g_{\theta,s}(y) \leq a^{-d/2} \varphi_1(y), \\ &p_{\theta,s}(y) \geq c_0 \varphi_a(y), \qquad\qquad c_0 = 2^{-d} a^{d/2}. \end{aligned} \tag{8.2}$$

We next give a pointwise derivative bound, without a local-information approximation. Let J be the skew-symmetric rotation generator, with $\|J\|_{\mathrm{op}} = 1$. Then $R'_\theta = R_\theta J$, and

$$\partial_\theta \log g_{\theta,s}(y) = -\frac{1}{2} y^\mathsf{T} R_\theta [J, D_s^{-1}] R_\theta^\mathsf{T} y,$$

where $[A, B] = AB - BA$. For $D_s = I$, the commutator is zero. For D_1 above,

$$\left\| D_1^{-1} - I \right\|_{\mathrm{op}} \leq \frac{dk}{1 - dk} \leq \frac{10d}{9} k.$$

Since $[J, I] = 0$, the absolute value of the log-density derivative is at most $C_d k \|y\|^2$. Also $\|\partial_\theta z\| \leq a^{-1/2} \|y\|$. Differentiate the finite product in (8.1), use boundedness of h', and bound its remaining factors by $3/2$. This gives

$$|\partial_\theta H_{\theta,s}(y)| \leq C_d k \|y\| .$$

Constants may depend on A and the fixed bounds on h, h', and are uniform in k, s, θ. Together with (8.2), these inequalities imply

$$|\partial_\theta p_{\theta,s}(y)| \leq L_d k (1 + \|y\|^2) \varphi_1(y). \tag{8.3}$$

Integrating from θ' to θ and using the denominator bound in (8.2), we obtain

$$\begin{aligned}\chi^2(P_{\theta,s}\|P_{\theta',s}) &= \int \frac{(p_{\theta,s}-p_{\theta',s})^2}{p_{\theta',s}} \\ &\le \frac{L_d^2}{c_0}k^2|\theta-\theta'|^2\int(1+\|y\|^2)^2\frac{\varphi_1(y)^2}{\varphi_a(y)}\,dy \\ &\le L_*^2k^2|\theta-\theta'|^2. \end{aligned} \tag{8.4}$$

The integral is finite: apart from a polynomial and a fixed normalizing constant, its exponential factor is

$$\exp\left[-\left(1-\frac{1}{2a}\right)\|y\|^2\right] = \exp\left[-\frac{4}{9}\|y\|^2\right].$$

This also justifies the differentiations and integrations. For $S=0$, the Gaussian derivative vanishes and the product multiplier alone gives the same bound. At $k=0$, all regimes are Gaussian and isotropic, so every rotation has the same observation law.

Let P_θ^T be the full independent sample law with the given regime counts. The inequality $\log x \le x-1$ gives $\mathrm{KL}(P\|Q)\le\chi^2(P\|Q)$. Additivity of log likelihood ratios for independent observations yields

$$\mathrm{KL}(P_\theta^T\|P_{\theta'}^T) \le TL_*^2k^2|\theta-\theta'|^2. \tag{8.5}$$

We may use $\mathrm{TV}(P,Q)\le\sqrt{\mathrm{KL}(P\|Q)}$. For a direct verification, put $a_0=\int\sqrt{pq}$. Jensen's inequality gives

$$\mathrm{KL}(P\|Q)\ge -2\log a_0 \ge 2(1-a_0) = \int(\sqrt{p}-\sqrt{q})^2.$$

Cauchy–Schwarz applied to $(\sqrt{p}-\sqrt{q})(\sqrt{p}+\sqrt{q})$ gives $\mathrm{TV}(P,Q)^2\le 2(1-a_0)$. Hence

$$\mathrm{TV}(P_\theta^T,P_{\theta'}^T)\le L_*k\sqrt{T}\,|\theta-\theta'|. \tag{8.6}$$

Adding the same independent randomization device to both experiments leaves their total variation distance unchanged. The lower bounds therefore apply to randomized confidence procedures.

9 Matching confidence lower bounds at every confidence level

The rotation targets have a uniformly nondegenerate metric on the short arc. For $|\theta|,|\theta'|\le\theta_0=1/8$,

$$\|R_\theta-R_{\theta'}\|_{\mathrm{F}} = 2\sqrt{2}\left|\sin\frac{\theta-\theta'}{2}\right|, \qquad \|R_\theta-I\|_{\mathrm{F}}\le\sqrt{2}\,\theta_0.$$

Every nonidentity signed permutation P has $\|I-P\|_{\mathrm{F}}\ge 2$: either a diagonal sign changes, or at least two diagonal entries vanish and their off-diagonal entries contribute the corresponding squared norm. The triangle inequality gives

$$\|R_\theta-R_{\theta'}P\|_{\mathrm{F}}\ge 2-2\sqrt{2}\,\theta_0 \quad (P\ne I),$$

whereas the identity matching has norm at most $2\sqrt{2}\,\theta_0$. Thus the identity is the closest matching on this arc. Using $\sin x\ge 2x/\pi$ for $0\le x\le\pi/2$, we obtain

$$d_{\mathcal{G}}([B_\theta],[B_{\theta'}])\ge c_{\mathrm{rot}}|\theta-\theta'|, \qquad c_{\mathrm{rot}}=\frac{2\sqrt{2}\,b_0}{\pi}>0. \tag{9.1}$$

Here π denotes the circle constant.

Fix $0 < \alpha < 1$, put $p = 1 - \alpha$, and enlarge L_* to be positive if necessary. Define

$$\eta_T = \begin{cases} \min\{\theta_0, p/(4L_* k\sqrt{T})\}, & k > 0, \\ \theta_0, & k = 0. \end{cases}$$

Every experiment with $|\theta| \le \eta_T$ belongs to $\mathcal{M}_T(k)$, and (8.6) gives $\mathrm{TV}(P_\theta^T, P_0^T) \le p/4$. At $k = 0$, this follows from exact equality of the laws. Let C be any finite-T honest procedure in the definition of $\mathcal{R}_T(k, \alpha)$. For each such θ,

$$\mathbb{P}_0^T\{[B_\theta] \in C\} \ge p - \frac{p}{4} = q, \qquad q = \frac{3p}{4}. \tag{9.2}$$

Choose an integer $N \ge 2$ with $Nq \ge 2$, and take N equally spaced angles in $[-\eta_T, \eta_T]$, with spacing $\Delta_\theta = 2\eta_T/(N-1)$. Let Z count the targets contained in C. Then $\mathbb{E}_0 Z \ge Nq$. If $Z \ge 1$, the largest and smallest covered indices differ by at least $Z - 1$; (9.1) gives

$$\mathrm{diam}(C) \ge c_{\mathrm{rot}} \Delta_\theta (Z-1)_+.$$

This is also true when C is empty. Since $(Z-1)_+ \ge Z - 1$,

$$\mathbb{E}_0 \,\mathrm{diam}(C) \ge c_{\mathrm{rot}}\, 2\eta_T \frac{Nq - 1}{N - 1} \ge c_{\mathrm{rot}} q \eta_T. \tag{9.3}$$

The last inequality uses $Nq \ge 2$ and $N/(N-1) \ge 1$. Only finitely many measurable coverage events are used; no joint measurability in a continuum parameter is presumed.

For positive A, B and $x \ge 0$, $\min\{A, Bx\} \ge \min\{A, B\} \min\{1, x\}$. Hence, also at $k = 0$,

$$\eta_T \ge \min\left\{\theta_0, \frac{1-\alpha}{4L_*}\right\} r(T, k).$$

The reference model $\theta = 0$ belongs to $\mathcal{M}_T(k)$, so its expected diameter bounds the supremum risk from below. Taking the infimum over honest procedures proves the lower half of Theorem 1, with

$$c_\alpha = c_{\mathrm{rot}} \frac{3(1-\alpha)}{4} \min\left\{\theta_0, \frac{1-\alpha}{4L_*}\right\} > 0.$$

This holds for every $0 < \alpha < 1$; no restriction $\alpha < 1/2$ was used. Combining with Section 6 proves (1.3).

For asymptotic honesty, set $\alpha' = (1+\alpha)/2$. The defining lower limit implies that every asymptotically honest sequence eventually has coverage at least $1 - \alpha'$, uniformly over $\mathcal{M}_T(k_T)$. Apply the finite-T argument with α' at each such T. It gives (1.4) with $c'_\alpha = c_{\alpha'} > 0$. The finite-sample procedures of Sections 3 and 6A are automatically asymptotically honest. This proves the separate asymptotic assertion, including arbitrary weakening rates and the identically zero sequence. □

10 Sharp squared-error point estimation

The same geometry gives a second informativeness criterion. We do not infer squared-error risk from coverage at a fixed confidence level. Let Θ_0 be the compact outer set without a strength restriction and choose a fixed countable dense sequence (ϑ_j) in it. The function

$$d_F(y) = \inf_{\vartheta \in \Theta_0} \|F(\vartheta) - y\|$$

is the distance to the compact set $F(\Theta_0)$, so it is continuous and its infimum is attained. Choose the first index j whose residual at $\widehat{F}$ is at most $d_F(\widehat{F}) + T^{-1}$. Such an index exists by density and

continuity, and the first-index selection is Borel. Let $\widehat{B}$ be its B component. This is an explicit measurable approximate minimum-distance rule; it does not assume a measurable choice of an unspecified exact optimizer.

For a true tuple ϑ_*, write $e_* = \left\|\widehat{F} - F(\vartheta_*)\right\|$. Since $\vartheta_* \in \Theta_0$,

$$\|F(\vartheta_j) - F(\vartheta_*)\| \le 2e_* + T^{-1}.$$

Use (6.1) with the true tuple first. For $\delta_* \ge k > 0$, $k \le \bar{k} \le 1$,

$$d_{\mathcal{G}}([\widehat{B}], [B_*])^2 \le \min\left\{ D_{\max}^2, \left(\frac{L_I}{k}\right)^2 (2e_* + T^{-1})^2 \right\}.$$

The inequality $(x+y)^2 \le 2x^2 + 2y^2$ and (2.1) give

$$\mathbb{E}_m d_{\mathcal{G}}([\widehat{B}], [B_*])^2 \le \min\left\{ D_{\max}^2, \left(\frac{L_I}{k}\right)^2 \left(\frac{8V}{T} + \frac{2}{T^2}\right) \right\}.$$

Thus the worst-case squared-error risk is at most a fixed multiple of $\min\{1, (Tk^2)^{-1}\}$. At $k = 0$, boundedness of the target gives a constant upper bound. The estimator itself does not use k.

For the matching lower bound, take the rotation models with angles $\pm\eta_T^{\mathrm{pt}}$, where

$$\eta_T^{\mathrm{pt}} = \begin{cases} \min\{\theta_0, (4L_* k\sqrt{T})^{-1}\}, & k > 0, \\ \theta_0, & k = 0. \end{cases}$$

Their total variation distance is at most $1/2$ by (8.6), and their target separation Δ satisfies $\Delta \ge 2c_{\mathrm{rot}}\eta_T^{\mathrm{pt}}$ by (9.1). An estimator yields a test by assigning its estimate to the closer target. On a test error, the estimation distance from the true target is at least $\Delta/2$, by the metric triangle inequality. The sum of test-error probabilities is at least $1 - \mathrm{TV}$: for the corresponding measurable event E,

$$P_0(E) + P_1(E^c) = 1 + P_0(E) - P_1(E) \ge 1 - \mathrm{TV}(P_0, P_1).$$

The larger squared-error risk is consequently at least

$$\frac{\Delta^2(1 - \mathrm{TV})}{8} \ge \frac{\Delta^2}{16}.$$

This is a positive constant times $\min\{1, (Tk^2)^{-1}\}$, also at $k = 0$. The randomization observation in Section 8 makes the bound valid for randomized estimators. Hence the approximate minimum-distance rule has the sharp worst-case squared-error rate in the same declared experiment.

11 Explicit labels for a labelled impact matrix

The quotient convention describes what the statistical features identify without additional economic labels. A labelled response needs an actual label restriction; sorting nearly equal estimates would not give a uniform label guarantee. Here is a precise nonempty alternative class. With $b_0 = (b_- + b_+)/2$, fix

$$0 < \rho \le \min\left\{ \frac{b_0}{4}, \frac{b_+ - b_-}{4} \right\},$$

and restrict B to the known closed region

$$\mathcal{K}_{\mathrm{lab}} = \{B : \|B - b_0 I\|_{\mathrm{F}} \le \rho\}.$$

This explicit loading and label restriction is imposed on both structural models and candidates. It says that the ordered columns load near the specified positive coordinate directions. It supplies statistical labels; attaching economic names still requires that the restrictions have the asserted economic interpretation. That interpretation is not inferred solely from the observation law.

All matrices in $\mathcal{K}_{\mathrm{lab}}$ have singular values between $b_0 - \rho$ and $b_0 + \rho$, inside the original spectral interval. For B, B' in this region, the identity matching has distance at most 2ρ. For every $P \neq I$ in $\mathcal{G}$,

$$\begin{aligned}\left\|B - B'P\right\|_{\mathrm{F}} &\geq b_0 \left\|I - P\right\|_{\mathrm{F}} - \left\|B - b_0 I\right\|_{\mathrm{F}} - \left\|B' - b_0 I\right\|_{\mathrm{F}} \\ &\geq 2b_0 - 2\rho > 2\rho.\end{aligned}$$

Thus $d_{\mathcal{G}}([B], [B']) = \|B - B'\|_{\mathrm{F}}$ throughout this region.

Intersect Θ_0 or Θ_k with this region before constructing confidence sets and estimators. Every concentration and inverse argument remains valid in ordinary Euclidean matrix distance. The lower family stays in the region after replacing θ_0 by

$$\min\left\{\frac{1}{8}, \frac{\rho}{2b_0}\right\},$$

since $\|b_0 R_\theta - b_0 I\|_{\mathrm{F}} \leq \sqrt{2}\, b_0 |\theta| \leq \rho$. Therefore the same sharp confidence-diameter and squared-error rates hold for the labelled impact matrix on this explicit class, at every confidence level and for every weakening sequence. This corollary does not assert that the general unlabelled experiment identifies economically named shocks.

12 Response functionals and strongly identified components

The primary response in Theorem 1 is the full quotient impact matrix; Section 11 gives a labelled full-matrix response. For a continuous response map ψ on either declared compact target, the image of a constructed confidence set has coverage for $\psi(B)$. Its image is compact and is measurable as a random closed set: its hit event for an open set is the hit event of the original set for the open preimage. If ψ has Lipschitz constant L_ψ, then

$$\operatorname{diam}(\psi(C_T)) \leq L_\psi \operatorname{diam}(C_T).$$

It inherits the expected-diameter upper rate and the point-estimation upper rate, the latter with squared constant L_ψ^2. More generally, the image diameter is bounded pointwise by $\omega_\psi(\operatorname{diam}(C_T))$, where ω_ψ is the uniform modulus of continuity on the compact target. An upper bound alone implies no matching lower rate.

A precise sufficient condition for the matching lower rate is a nonflat weak-rotation response: on a short hard arc above, suppose

$$\|\psi([B_\theta]) - \psi([B_{\theta'}])\| \geq \ell_\psi |\theta - \theta'| \qquad (\ell_\psi > 0)$$

for every pair of angles in the arc. The finite-grid proof in Section 9 applies with ℓ_ψ in place of c_{rot}, as does the two-point proof in Section 10. These lower rates match the preceding Lipschitz upper rates for responses satisfying both conditions. For labelled matrices an off-diagonal impact coordinate is an example: $(B_\theta)_{12} = -b_0 \sin\theta$, with derivative bounded away from zero on a sufficiently short arc. A constant response has zero estimation risk and zero confidence diameter and does not satisfy the nonflat condition.

A distinct response is $\Sigma_0 = BB^{\mathsf{T}}$, identified by the baseline second moment without resolving the rotation. The confidence set

$$C_\Sigma = \left\{\Sigma : b_-^2 I \leq \Sigma \leq b_+^2 I, \quad \left\|\Sigma - \widehat{\Sigma}_0\right\|_{\mathrm{F}} \leq R_T\right\}$$

has coverage at least $1-\alpha$ by the same moment bound and has deterministic diameter at most $2R_T$. Approximate minimum distance over the deterministic compact spectral set $\{\Sigma = \Sigma^\mathsf{T} : b_-^2 I \le \Sigma \le b_+^2 I\}$, rather than over the possibly empty data-dependent confidence set C_Σ, gives squared-error risk $O(T^{-1})$, by the residual argument of Section 10. These bounds are uniform even at $\delta = 0$.

These covariance rates are sharp in the declared classes. For an explicit lower bound, retain the shocks of Section 7 and replace B_θ by

$$B_t = b_0 e^t I, \qquad |t| \le t_0,$$

where $t_0 > 0$ is sufficiently small that the singular values stay strictly between b_- and b_+. Shrink t_0 if necessary so that $e^{2t} \in [0.95, 1.05]$. For the labelled class, shrink t_0 further until

$$\sqrt{d}\, b_0 |e^t - 1| \le \rho \qquad \text{throughout } |t| \le t_0.$$

The scale family then stays in $\mathcal{K}_{\mathrm{lab}}$ as well. In coordinates divided by b_0, its density is

$$p_t(y) = e^{-dt} p_0(e^{-t} y).$$

Here p_0 is the appropriate regime density with covariance eigenvalues in $[0.9, 1]$ and multiplier between 2^{-d} and $(3/2)^d$. Its derivatives are bounded by a fixed polynomial times its Gaussian envelope, uniformly in $k \le \bar{k}$, because h, h' are bounded. Differentiation gives

$$\partial_t p_t(y) = -d p_t(y) - y \cdot \nabla_y p_t(y).$$

These derivatives have uniformly bounded weighted L^2 norm with inverse $N(0, 0.8I)$ density: their Gaussian covariance eigenvalues lie in $[0.855, 1.05]$, the density dominates a positive multiple of the $N(0, 0.8I)$ density, and $1.05 < 2(0.8)$ ensures integrability of the squared Gaussian envelope times any fixed polynomial. The mean-value and denominator argument of Section 8 therefore proves

$$\chi^2(P_t \| P_{t'}) \le C|t - t'|^2$$

in every regime. The T-sample total variation is at most $C\sqrt{T}\,|t - t'|$. The covariance curve $b_0^2 e^{2t} I$ has Frobenius distance bounded below by a positive constant times $|t - t'|$ on this interval. Apply the finite-grid proof on an interval of radius of order $T^{-1/2}$, and the two-point proof to two such parameters. They give, respectively, expected diameter at least $c_\alpha T^{-1/2}$ and squared-error risk at least c/T. The nuisance features remain those of Section 7 and meet $\delta \ge k$. These are lower bounds inside $\mathcal{M}_T(k)$.

Every Lipschitz response depending only on Σ_0 inherits the strong upper rate. A matching lower rate for a particular response requires a corresponding nonflat covariance curve. Rotation-sensitive, covariance-only, and flat responses must therefore be distinguished; the theorem does not assign the weak-orientation rate to every possible ψ.

13 Dimension one and two checks of the joint index

When $d = 1$, there is no rotational identification problem. Choose the column sign explicitly by $B > 0$. In the baseline regime, $\mathbb{E}E_0^2 = 1$, so $\mathbb{E}U^2 = B^2$. The baseline second-moment confidence interval and

$$|B - B'| \le \frac{|B^2 - (B')^2|}{2b_-}$$

give finite-sample uniform coverage and diameter $O(T^{-1/2})$ for B throughout the same bounded-moment class. Approximate minimum distance gives squared-error risk $O(T^{-1})$. For the lower

bound, choose Gaussian shocks in every regime, $D_s = 1$, and $B = b_0 e^t$ in an interior interval. Their eighth moments equal 105, so they are permitted. The scale-density calculation in Section 12, together with the finite-grid and two-point arguments, gives matching lower bounds for all confidence levels. The pairwise index δ is neither defined nor needed in dimension one; no one-shock weak-rotation rate is asserted.

The joint index can be positive even when neither feature collection identifies on its own. For $d = 3$, $S \geq 1$, and small $k > 0$, take

$$c = (0, k, 0), \qquad D_1 = \mathrm{diag}(1, 1, 1 - k),$$

and $D_s = I$ for all other regimes. Use density $f_{k/A}$ for shock 2, Gaussian densities for shocks $1, 3$, and scale the third shock by $\sqrt{1-k}$ in regime 1. Every coordinate eighth moment is at most 105. The variance profiles of shocks $1, 2$ coincide, while the fourth cumulants of shocks $1, 3$ are both zero. Each of the separate moment maps thus leaves a rotation unresolved. The joint squared pair margins are nevertheless $k^2, k^2, 2k^2$, so $\delta = k$, and the global inverse bound applies. This example illustrates the arbitrary-d theorem.

Finally, zero fourth cumulant does not imply Gaussianity. A variable taking values $\sqrt{3}$ and $-\sqrt{3}$, each with probability $1/6$, and 0 with probability $2/3$, has mean zero, variance 1, fourth moment 3, sixth moment 9, and eighth moment 27. It is non-Gaussian although its fourth cumulant is zero. Such laws are allowed by the model class. The constant worst-case lower bound at $k = 0$ uses the rotationally indistinguishable Gaussian subfamily. It does not assert that every individual $\delta = 0$ law is unidentified from its full distribution, or that higher-order features could not improve inference at that law.

14 Scope

The canonical C32-80 question requires declarations of the structural class $\mathcal{M}_T$, observation and dependence model, moment assumptions, signs and labels, response, and informativeness criterion. Sections 1 and 11 supply these declarations. The core experiment allows arbitrary unknown conditional shock distributions under coordinate independence and an eighth-moment bound. Normal or parametric laws are used only as explicit lower-bound subfamilies. The full impact response with its sign and permutation convention is the primary response; the labelled and other-response results have their conditions stated separately.

Theorem 1 provides finite-sample uniform coverage, sharp worst-case expected diameter as the variance and fourth-feature lower bound shrinks, and the corresponding conclusion for the asymptotic coverage quantifier. Section 6A provides a single procedure that does not need to know that lower bound. Section 10 gives matching minimax squared-error bounds. The nonshrinking case and every sequence k_T are included, with $k, k_T \in [0, \bar{k}]$.

Known regime dates, observed innovation vectors, a common B, independence across observations, the stated eighth-moment bound, and fixed d are explicit parts of the declared experiment. Unknown regimes, changing B, estimated VAR residuals, heavy-tail classes without the required moments, and other non-Gaussian-feature indices are different models; no theorem for them is claimed without their own assumptions. A universal response-independent rate would also be false, as the constant and covariance responses demonstrate. The results address the fully declared structural-inference problem, rather than a classification of every model or every source of non-Gaussian information in the literature.

Lewis [1], especially Sections 5 and 7.2, supplies the identification-robust agenda and the normalization and label warnings.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] Daniel J. Lewis. Identification Based on Higher Moments in Macroeconometrics. *Annual Review of Economics* **17** (2025), 665–693. doi:10.1146/annurev-economics-070124-051419. Published article in the author's institutional repository. Agenda locators: Sections 5 and 7.2.